Waiting for Closure: Deep Networks Link Contours but Do Not Complete Them

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Abstract

Whether deep networks exhibit the Gestalt law of closure has been answered both ways, using stimuli that are rarely sized to the network's sampling grid and measures that are rarely checked against observers that cannot close contours. We built a patch-aware test battery in which every gap, inducer and dash is specified in units of the model's patch size and rendered natively, with same-ink controls that rotate, scramble or re-orient the pieces. We trained and tested 24 networks: frozen and fine-tuned CNNs and vision transformers, looped (weight-tied, recurrent-depth) transformers, and fGRU-H, γ-net and deep fGRU-H/fGRU circuits built from the published fGRU cell, including stacks of up to eight recurrent layers behind a CNN stem. Illusory contours are at best weakly represented: ViT-B/16 variants show a small but consistent advantage of Kanizsa figures over rotated-inducer controls (+0.08 to +0.14 across three fine-tuned seeds, +0.10 frozen), with Kanizsa accuracy of at most 0.39 (chance 0.10) against 1.00 on complete outlines; DINOv2-S shows an inconsistent advantage (+0.01 to +0.15 across seeds), and in the other networks it stays at or below +0.07, far from the fragment effects. Fine-tuned transformers do integrate fragmented outlines, and the arrangement carries the effect: scrambling piece positions while keeping orientations lowers accuracy by +0.46 to +0.65, and pieces placed correctly but misoriented are not recognized. Neither looping nor recurrence consistently adds closure: in five of six recurrent circuits a control trained with a single iteration matches or exceeds the eight-iteration network, although the recurrent states in the gap become more similar to those for the complete outline over iterations. Finally, we audited published closure measures with null observers. A blur reproduces the illusory-brightness signature reported for a predictive-coding network; a pixel-growth observer, which only sees how far ink lies from its centroid, scores +0.289 on the closure measure of Kim et al. [19], several times any CNN, and rotating each corner in place about its own ink centroid, which removes the radial difference this observer detects, reduces the measure to +0.005 and +0.005 in the two networks previously reported to show closure most consistently. The configural-effect (CE) measure survives null observers and is specific to closing arrangements, but it generally grows as the outline approaches completion, its advantage over a non-closing arrangement gains nothing further, on average, from a fully closed square, and changes with inducer shape and size at a fixed ratio of luminance-specified to total edge length, which predicts the clarity of human illusory boundaries across display sizes [30]. In 9 of 10 CNN and recurrent networks the mid-gap closure index halves at gaps of 6–50 px, inside the network's measured effective receptive field. Networks trained specifically to trace contours, including weight-tied looped transformers, link aligned pieces only across gaps a few pixels wider than in training and label almost none of the missing contour as contour. Deep networks group nearby, consistently oriented contour pieces; they do not close gaps, and illusory contours are at most weakly represented.

1 Introduction

Closure, the tendency to perceive an incomplete figure as a whole, is among the oldest Gestalt laws [18, 35] and a standard test of whether artificial vision organizes the world as people do [19, 39]. Studies of deep networks disagree. Similarity-based analyses report closure in ImageNet-trained CNNs [19, 39]; accuracy-based protocols report its absence [38]; classification images suggest networks rely on inducer orientation rather than illusory edges [4]; and recurrent predictive-coding networks have been reported to perceive Kanizsa squares [24]. Classifiers of regular polygons degraded at their corners or at the middle of each edge behave unlike human observers, who recover shapes with deleted midsegments but not with deleted vertices [6]: ResNets suffer more from edge than from corner degradation, and a ViT equally from both [33]. For vision-language models, whose image encoders are vision transformers, answers to classic illusions persist when the illusion-inducing elements are inverted, which Sun et al. [31] trace to different causes in different models, from recall of the canonical answer to limits of visual processing.

Two problems run through this literature. First, stimuli are drawn in pixels at arbitrary scales and then resized, so the size of a gap relative to a vision transformer's patches or a CNN's receptive field is uncontrolled. A six-pixel gap need not leave any 16-pixel patch empty, although it remains detectable within its patch; a forty-pixel gap may exceed every receptive field below the last stage. Second, measures are checked against some null controls, such as untrained networks [19], but not against simple observers that *cannot* close contours yet share the stimuli's low-level statistics, such as the radial distribution of ink. If a blur, a filter bank or an untrained network produces the same signature, the signature is not evidence of closure.

We address both problems. Our contributions are: (i) a patch-aware battery with geometry in patch units and same-ink controls (rotated, position-scrambled and orientation-swapped pieces, Kanizsa inducers, and dashed, noisy linking segments); (ii) a comparison of 24 networks spanning feed-forward, looped-transformer and horizontal/top-down recurrent architectures built from the published fGRU cell, each recurrent circuit but one with a matched single-iteration control; (iii) a null-observer audit of four published closure measures; and (iv) a faithfulness check of the saliency overlays commonly used to argue that networks "see" a shape.

2 Stimuli

Networks classify regular polygons with 3 to 12 sides, the ten-way task of Zhang et al. [38]. All lengths are set in units of the patch size P (16 px for ViT-B/16 and the CNNs, 14 px for DINOv2) and rendered natively at 224 px with 4× supersampling; images are never resized. The circumradius is $4.5P$ for every class, so overall size carries no class information. Training uses complete outlines with random rotation, scale (±15%), position ($\pm P$) and background polarity. The test battery (Fig. 1; about 38,000 images per patch size) contains: complete outlines; *mid-gap fragments*, with a gap of $g \in \{0.25, \ldots, 4\}P$ removed from the middle of each edge (the manipulation of Zhang et al. [38], but in absolute patch units; midsegment deletion in 6; edge degradation in 33); *corner-removed fragments*, in which contour within $g/2$ of every vertex is removed so that corner angles cannot be counted (vertex deletion in 6; corner degradation in 33); *Kanizsa figures* with inducer radius $r \in \{0.5, \ldots, 2\}P$; and *dashed links*, in which the Kanizsa gap contains a real dashed segment (spacing 0–$2P$, contrast 0–1, pixel noise only on the segment). Every fragment condition has a same-ink control, a disordered control in the sense of Kim et al. [19], in which each piece is rotated about a point inside it (a straight piece about its midpoint; a corner piece of a mid-gap fragment about the centroid of the triangle formed by its vertex and its two ends) by a fixed pseudo-random

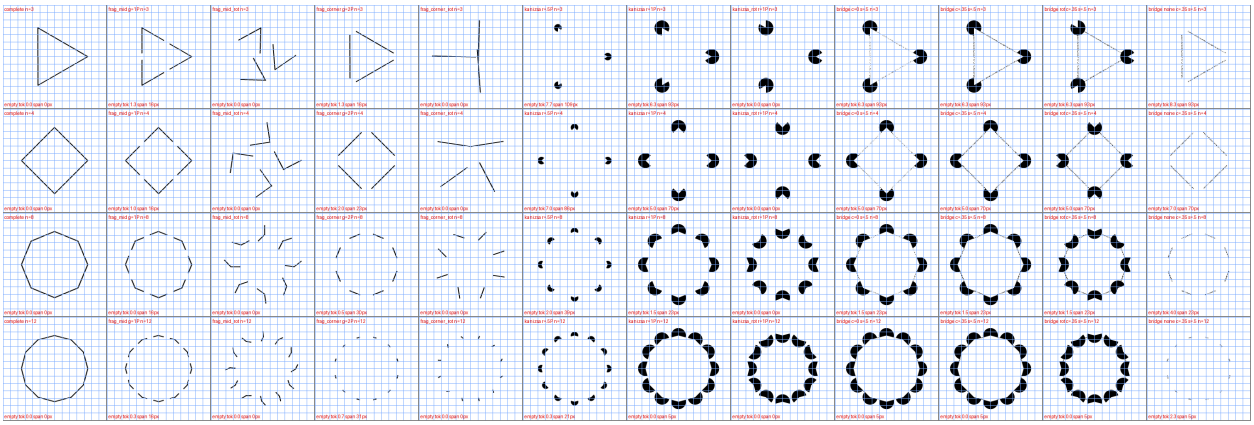


Figure 1: Patch-aware stimuli at 224 px, shown with the 16 px patch grid. Columns: complete outline; gaps in the middle of every edge and their rotated control; corners removed and their rotated control; Kanizsa figures with inducer radius 0.5 and 1 patch and the rotated-inducer control; dashed linking segments in the Kanizsa gap with inward, rotated and no inducers. Rows: 3, 4, 8 and 12 sides. Red text gives the number of ink-free patches each gap crosses.

angle (the rotated control). Each image is shifted by 0, $\frac{1}{4}$, $\frac{1}{2}$ or $\frac{3}{4}$ of a patch relative to the patch grid, and its geometry, gap masks and the number of ink-free patches each gap crosses are recorded.

Two further test-only controls separate closure from orientation statistics (Fig. 2B,C): *scrambled* stimuli move each piece to a random position inside the shape's disk while keeping its orientation and length; *cue conflict* stimuli, a position–orientation analogue of the shape–texture cue conflict of Geirhos et al. [14], keep each corner-removed piece's midpoint but give it the edge orientation of the polygon with one more side, so positions imply n sides and the orientations are n of the $n{+}1$ edge orientations of the polygon with one more side (for 12-sided polygons, the 11-sided one).

3 Models and protocol

Frozen backbones follow the protocol of Zhang et al. [38], who replace the last layer of a frozen network by a fully connected layer (Adam, 10^{-3}, 25 epochs): VGG16, ResNet50, and, not tested by them, ViT-B/16 and DINOv2-S/14. **Fine-tuned** ResNet50, ViT-B/16 and DINOv2-S/14 are trained end to end. **Looped transformers** split a pretrained ViT into prelude (blocks 0–3), a weight-tied core (blocks 4–7) applied T times, and coda (blocks 8–11), re-injecting the prelude output through a zero-initialized linear map so that $T{=}1$ is exactly the pretrained model (prelude, core and coda are the terms of 13; iterating a weight-tied block follows 10, 15); T is sampled from $\{1, \ldots, 4\}$ during training and swept to 12 at test. **Horizontal recurrence** uses the fGRU cell [20, 21] as specified in the appendix of Linsley et al. [21]: attention-modulated suppression gate, a gate that updates the recurrent state, channel-symmetric horizontal kernels, instance normalization of the gate inputs and the published initialization. Our adaptations are a VGG16 front end (conv2 features, stride 4), a convolutional readout (three stride-2 3×3 convolutions with ReLUs, global average pooling and a linear layer; a 1×1 convolution in the deep hierarchies and stacks below), unrolling for $T{=}8$, and, for the γ-net, two levels with max-pooled bottom-up connections. The models are a horizontal-only fGRU circuit (fGRU-H; the hGRU of Linsley et al. [20] is a different cell), an fGRU-H with a 1×1 readout that adds no spatial convolution after the recurrence, and a two-level γ-net whose top-down fGRUs update the lower layer from a projection of the upper one, read out from the lowest layer. **Deep fGRU-H/fGRU** hierarchies place fGRU layers at

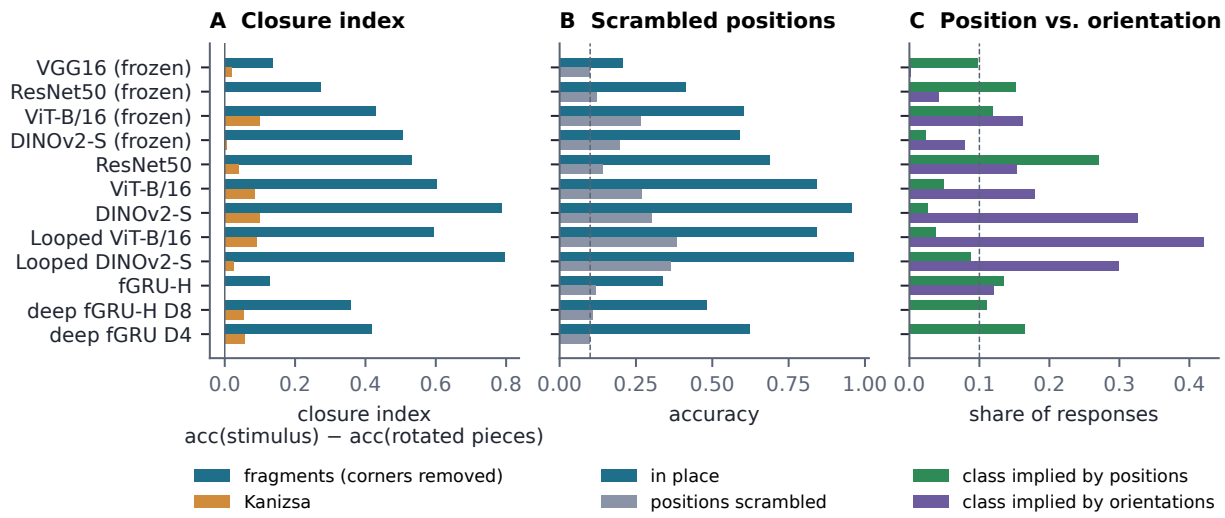


Figure 2: (A) Closure index for corner-removed fragments (all gap sizes) and Kanizsa figures. (B) Corner-removed fragments in place versus the same pieces with scrambled positions and unchanged orientations; dashed line, chance. (C) Cue-conflict stimuli: how often the network picks the class implied by the pieces' positions versus their orientations.

the four VGG stages: an eight-layer fGRU-H (two per stage, bottom-up only) and a four-layer fGRU with top-down connections, unrolled for T=8. **Divisive-normalization stacks** keep a CNN only at the input (VGG16 conv1–2, stride 4) and then stack 4 or 8 fGRU layers at that single resolution, each preceded by divisive normalization [16] with learned channel weights, as in generalized divisive normalization [5], over a fixed 5×5 average pool of our choosing; layers are linked by 1×1 convolutions, so spatial integration beyond the stem comes from the recurrent horizontal kernels, the 3×3 attention kernel of the suppression gate and the divisive-normalization pool, and top-down fGRUs update each layer through a zero-initialized gate. Every recurrent model except the fGRU-H with a 1×1 readout has a matched control trained with T=1. All networks except the frozen ones are trained for 15–20 epochs on 6,000 complete polygons and reach 1.00 validation accuracy; frozen backbones reach 0.97–1.00. Transformer inputs are normalized with the statistics of each checkpoint's own pretraining.

Measures. The *closure index* is accuracy on a stimulus minus accuracy on its same-ink rotated control. We also report the closure measure $\bar{C}$ of Kim et al. [19] on our stimuli (cosine similarity to the matched complete image, stimulus minus control), the linear probe of Zhang et al. [38] trained at every layer [a linear classifier probe; 3], the effective receptive field [22], and, for recurrent models, the *gap similarity*, $\bar{C}$ restricted to the gap region of the recurrent state (for Fig. 8B: one-patch corner-removed fragments, all cells the missing contour crosses, the first recurrent layer; the gap analysis uses mid-gap fragments, cells empty in both images and the layer the classifier reads): $\cos(h_t^{\text{stim}}, h_t^{\text{complete}}) - \cos(h_t^{\text{ctrl}}, h_t^{\text{complete}})$ at each iteration t.

4 Results

4.1 Illusory contours are at most weakly represented

Across all networks that learned the training task (first training seed of each), accuracy on Kanizsa figures lies between 0.04 and 0.39 (chance 0.10), and the Kanizsa closure index reaches at most

+0.10 (ViT-B/16 (frozen); Fig. 2A). Across three to five training seeds of three networks (Table 3), fragment closure indices and arrangement effects (accuracy in place minus accuracy with scrambled positions; see below) replicate (largest s.d. 0.03 for corner-removed fragments and 0.05 for the arrangement effect); the Kanizsa index is small but consistent in fine-tuned ViT-B/16 (+0.08 to +0.14, Kanizsa accuracy 0.18–0.35), inconsistent in DINOv2-S (+0.01 to +0.15) and near zero in ResNet50 (-0.05 to +0.07), always against 1.00 accuracy on complete outlines (Table 3). The frozen and looped ViT-B/16 show indices of the same size (Table 1), so ViT-B/16 carries a weak illusory-contour signal that is consistent across seeds only in this network. Class-specific relevance maps [8] for the top-1 class illustrate this: in Fig. 7 Kanizsa triangles and squares are labelled as hexagons, and their rotated inducers receive the same label. Dashed linking segments raise accuracy about as much with aligned inducers as with no inducers: from contrast 0 to 1 the median gain over 22 networks is +0.14 with aligned inducers and +0.11 without inducers (difference -0.01; above +0.05 in 6 networks and below −0.05 in 8), whereas rotated inducers suppress it (+0.01). Noise was drawn independently for each image.

4.2 Fine-tuned transformers integrate fragments, and the arrangement matters

With corners removed, fine-tuned ViT-B and DINOv2 (plain and looped) classify fragmented outlines at 0.84–0.96 in the orientation-control battery (a separate test set of corner-removed fragments in place, scrambled and cue conflict, gaps of 0.5–2P), and their closure index over all gap sizes of the main battery is +0.60 to +0.80. Frozen backbones reach half to 70% of that, and fine-tuning raises the index 1.4–2× within each family. Scrambling the pieces' positions while keeping their orientations drops the fine-tuned transformers to 0.27–0.39 (Fig. 2B): the arrangement adds +0.46 to +0.65 beyond orientation statistics. When positions and orientations conflict, these networks pick the class implied by positions only 0.03–0.09 of the time and the class implied by orientations 0.18–0.42 (Fig. 2C). Pieces must therefore be both placed and oriented along the contour, the signature of contour integration by good continuation [12] rather than of position or orientation alone.

The manipulation used by Zhang et al. [38], which keeps the corners, behaves differently: scrambling corner pieces costs fine-tuned transformers only +0.12 to +0.17, so that test is largely solved from corner angles and cannot separate closure from corner reading.

4.3 Patch geometry decides who can bridge a gap

Plotted against the number of ink-free patches a gap crosses, accuracy separates the network families (Fig. 3); we did not test whether this count predicts accuracy better than gap size. Averaging, for 3–6-sided polygons, over gaps that cross three or more empty patches (bins of 3, 4, 5 and 6 or more empty patches weighted equally), fine-tuned transformers score 0.78–0.99, while the CNNs and the representative fGRU-H/fGRU circuits score 0.10–0.39.

The bridgeable gap is far smaller than the receptive field. For each network we define a critical gap g^*, the gap at which the mid-gap closure index falls to half its value at the smallest gap, and compare it with the measured ERF_{50} for CNNs and recurrent circuits (our summary of the effective receptive field of Luo et al. [22]: the radius around the gradient centroid that contains half of the absolute gradient of a central unit with respect to white-noise input) and with the patch size for transformers (Figs. 4, 5). A transformer's attention can span the whole image, and the analogue of receptive-field size proposed by Dosovitskiy et al. [11], the mean attention distance, varies across heads and layers; we therefore use the patch size, the resolution at which the gap is tokenized, as the reference. We compute the index on the fixed set of 3- to 6-sided polygons, for which every gap size

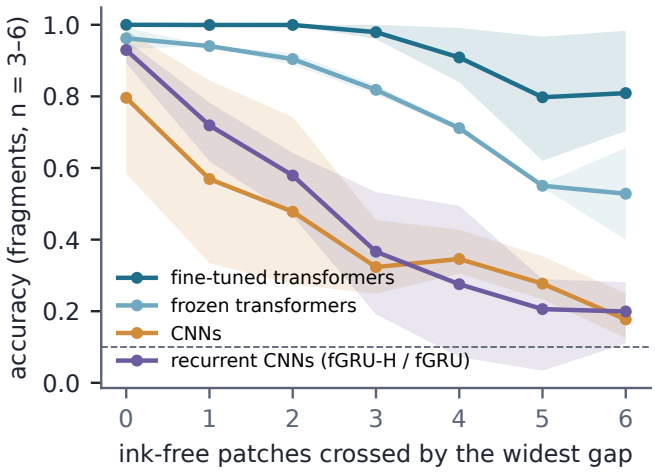


Figure 3: Accuracy on fragments ($n = 3$–6, all gap sizes valid) against the largest number of ink-free patches crossed by any gap. Lines: group means; bands: range across networks.

is valid; pooling all classes would change the mix of classes with gap size, because larger gaps exceed the edges of many-sided polygons. For CNN and recurrent networks other than the single-iteration controls, g^* lies between 6 and 50 px; it marks where the index halves, and a smaller residual effect can persist beyond it. For 9 of 10 CNN and recurrent networks it is smaller than ERF_{50} (median ratio 0.44; 14 of 16 including the single-iteration controls), and its rank correlation with ERF_{50} is only +0.33; the exception, excluding single-iteration controls, is the fGRU-H with a 1×1 readout, whose readout adds no spatial convolution after the recurrence and which has the largest fragment closure index of any recurrent circuit (+0.63). In 9 of 10 CNN and recurrent networks (excluding single-iteration controls) the index therefore halves at gaps whose full length fits inside the measured ERF_{50}, a radius; a smaller effect can remain at larger gaps. ERF_{50} is measured on noise images, and for the deep hierarchies at the first recurrent layer, so this comparison is descriptive: it shows that the gap lies within the region that influences a unit, not that the network uses that region for the task. Transformers behave differently: 3 of 6 keep more than half their mid-gap index up to the largest gap tested (four patches), and the others halve at 3.4–3.6 patches. Mid-gap fragments keep the corners, which these networks read directly, so this index overstates how far they bridge gaps.

4.4 Where closure emerges, and what the maps show

Applying the linear probe at every layer (Fig. 6), fragment closure in fine-tuned transformers rises through the middle blocks and peaks late (ViT-B +0.68, DINOv2-S +0.79). In the looped ViT it reaches +0.73 by the fourth pass through the shared core and gains at most about 0.02 with further iterations, so additional loops add depth without adding closure. Frozen ViT-B peaks at +0.60 and fine-tuned ResNet50 at +0.52 in its last stage. Relevance maps make the dissociation between fragments and illusory contours visible (Fig. 7): for fragmented outlines, relevance follows the whole contour including its gaps, whereas for Kanizsa figures it stays on the inducers, and Kanizsa triangles and squares are labelled “hexagon” whether their inducers face inward or are rotated.

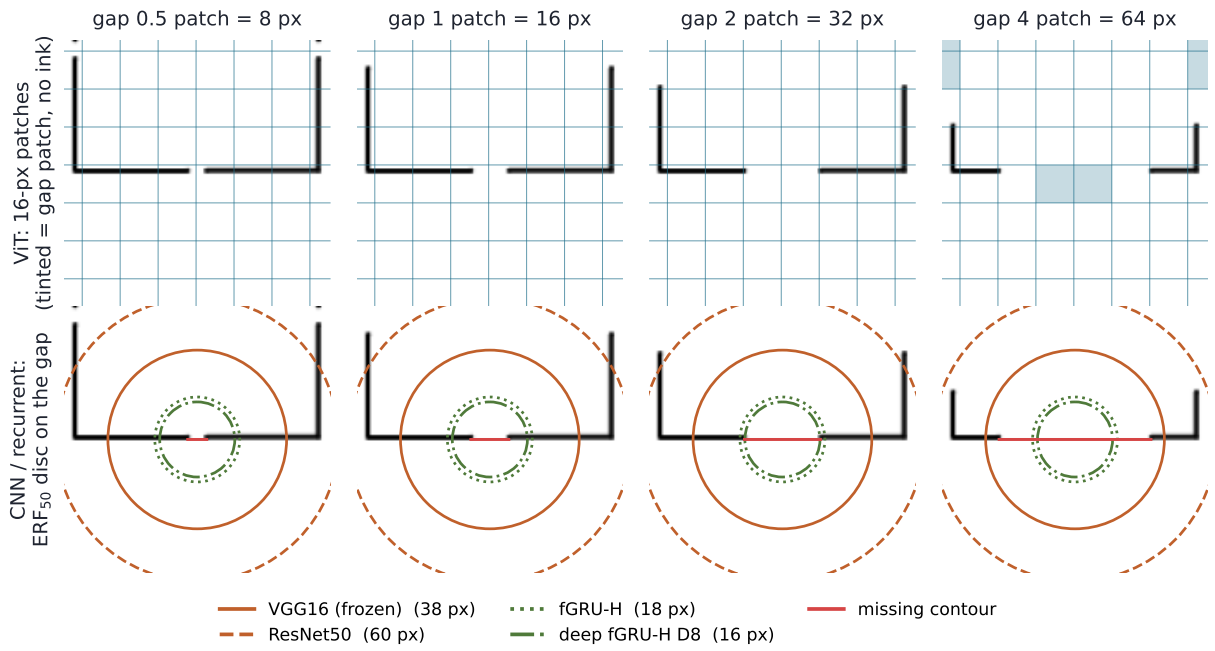


Figure 4: The same mid-gap fragment at four gap sizes, zoomed on one gap. Top: the 16 px patch grid a ViT-B/16 tokenizes; tinted patches are crossed by the missing contour and contain no ink. Bottom: the measured 50% effective receptive field (ERF_{50}) of four networks, centred on the gap (red: missing contour). Even the 64 px gap lies inside the ERF_{50} of two of the four networks shown.

4.5 Recurrence does not consistently add closure

Looping a fine-tuned transformer changes the fragment closure index by -0.01 to +0.01. **Recirculation.** Recirculation [23] turns a frozen transformer into a dynamical system by leaking a renormalized deep-layer activation back into a shallow layer at every step. On a single static image there is no step dimension, so what we test is training-free recurrence in depth (looping, in the terms of 23) with recirculation's mixing rule, in which the source activation is rescaled to the destination's norm before being added: fine-tuned and frozen ViT-B/16 and DINOv2-S re-process the same image with the previous pass's deep state mixed into the shallow layer, over 60 settings per network of source layer, destination layer, mixture coefficient and number of passes. Among settings that lowered accuracy on complete outlines by at most five percentage points, none raised the Kanizsa closure index by more than +0.03 or the fragment index by more than +0.03 (best Kanizsa index of a fine-tuned network +0.14), small differences, selected post hoc, whose reliability we did not estimate. For fGRU-H and fGRU circuits, recurrence increases the gap similarity: it rises from 0.28 to 0.35 (fGRU-H), 0.17 to 0.42 (γ-net) and 0.35 to 0.37 (eight-layer deep fGRU-H) over iterations (Fig. 8B). Yet increased gap similarity does not consistently accompany a higher closure index. Recurrence does not consistently improve the fragment closure index: in 5 of 6 circuits the control trained with a single iteration is within 0.02 of, or above, its eight-iteration counterpart (for example the fGRU-H, +0.13 versus +0.18; Fig. 8A), and the one gain, +0.08 in the deep fGRU D4, is from a single seed. **Divisive normalization with a CNN stem only.** When the CNN is reduced to a stem and every further layer is a recurrent fGRU preceded by divisive normalization, the corner-removed closure index is +0.29 for the four-layer and +0.26 for the eight-layer stack, well below the fine-tuned transformers; the single-iteration controls score +0.30 and +0.40, and the Kanizsa index of all 4 stacks stays at or below +0.01. Across recurrent circuits the Kanizsa index is

at most +0.06 (single-iteration controls +0.04), so recurrence adds no illusory-contour signal either.

4.6 Networks trained to trace contours link pieces but do not complete them

Our recurrent circuits were trained on complete outlines, a task that never demands long-range integration. We therefore probed networks trained for exactly that: hGRU, xhGRU (an hGRU whose per-iteration batch normalization is replaced by group normalization [36] shared across iterations), fGRU/γ-net and weight-tied looped local-attention transformers trained to segment the dashed contour on which a marker sits, a Pathfinder-style curve-tracing task [17, 26] (validation IoU 0.92–0.996 on 128 px images; taken from a companion project without retraining; the training target is the full dashed contour, so a network that completes gaps has no reason to label them). Each saw a marker on a short dashed segment followed, along the same line or arc, by a second segment after an extra gap, or by the same segment rotated in place (segment-pair probes; Fig. 9). *Traced* is the share of the target's ink labelled as part of the marked contour, and *gap coverage* the share of the missing contour labelled as contour (both with probability above 0.5). At the dash spacing of their training data, the tied transformer, the hGRU and the two xhGRUs extended the trace onto the aligned segment (mean 0.55–0.75 of its ink) and almost never onto the rotated one (mean at most 0.06): they link by good continuation. For these four tracers, pooling aligned straight and curved segment pairs, linking halved after only 5–6 px of extra gap, averaged 0.02 at 12 px, and mean coverage of the missing contour stayed below 1% of gap pixels (at most 0.01; the fGRU reaches 1.5%). Running the looped transformer for twice its trained iterations reduced tracing of the aligned target at native spacing (0.55 to 0.41) and did not extend linking across larger gaps, an untied 24-block transformer of the same depth traced only 0.08 of the aligned target at native spacing, and the fGRU/γ-net labelled the rotated segment nearly as often as the aligned one (0.75 versus 0.82), so its spread is not contour-specific. In their segmentation output, networks that solve contour tracing therefore show the same profile as our classifiers: linking of present, aligned ink over short distances, and almost no labelling of absent contour; their training target did not require such labelling, so this does not rule out completion in their internal states.

4.7 Auditing published closure measures

We recreated each measure's stimuli from its published specification and computed it on trained networks and on null observers that cannot close contours: untrained networks (three seeds for the closure-measure audit, one for the extended configural-effect controls; the brightness audit uses no networks), a Gabor energy bank, raw pixels, and a *pixel-growth* observer that encodes only the radial distribution of ink about its centroid (our term; it differs from the pixel overlap that Kim et al. [19] control, which compares images pixel by pixel; Fig. 10).

Illusory brightness [24]. The reported evidence is a figure-ground luminance difference (FG) of about 0.02 in reconstructions of Kanizsa squares whose inducers all face inward (All-in), which stays above zero after the first time step. Diffusing the input image produces an FG of +0.016 for All-in and -0.003 and -0.002 for the All-out and Random conditions after 25 steps, rising and then decaying as reported for their predictive-coding VGG (PVGG). A PCA reconstruction learned from natural images, with no feedback, gives +0.055 versus -0.032. The FG signature therefore does not require contour completion; it follows from smoothing inducers whose mass lies outside the illusory square.

Closure measure [19, 39]. The closure measure $\bar{C}$ is the cosine similarity of a network's representations of aligned fragments and a complete triangle minus that of disordered fragments (corners rotated about their vertices, as in 39) and the complete triangle. Our replication reproduces

the published pattern: trained CNNs score +0.019 to +0.087, rising with edge length, and untrained networks score +0.003. But the pixel-growth observer, which discards all angular information, scores +0.289 (+0.452 at the longest edges), several times any network. The disordered control rotates each corner about its vertex and pushes ink outward; when we instead rotate each corner in place about its own ink centroid, VGG16 falls from +0.029 to +0.005 and DenseNet-121 from +0.025 to +0.005, the two networks reported to show closure most consistently [39]. A residual of +0.013 to +0.028 remains in 7 of the 9 networks. The pairing rule also matters: Kim et al. [19] pair complete and fragment images that differ in global orientation and position, to ensure that there is no more pixel overlap between complete and aligned images than between complete and disordered images; when the paired images share orientation and position, raw pixels alone score +0.200 (versus +0.011 with the published pairing), and the published version of Zhang et al. [39] does not state which pairing was used. Averaged over edge lengths, every trained network scores at most +0.014 on Kanizsa triangles, and ViT-B/16 scores +0.066 on line fragments.

Configural effect [7, 39]. The network configural effect compares the distance between the representations of a base pair of images with that of a composite pair formed by adding the same components, the context, to both: $\mathrm{CE} = (D_{\mathrm{composite}} - D_{\mathrm{base}})/(D_{\mathrm{composite}} + D_{\mathrm{base}})$ with D the Euclidean distance between representations, positive for a configural superiority effect and negative for a configural inferiority effect. In the line-segment condition of Zhang et al. [39] the context closes a square; we call it the closing context. This measure survives the null observers. Pixels, Gabor filters and untrained networks score near zero, and with components long enough that the gaps are under half a patch, ResNet50 and DenseNet-121 reach +0.30 and +0.29. Replacing the context with the same components rotated outward, so that no closed shape can form, reverses the effect (-0.17 and -0.18). Three further controls show what it measures (stimuli in Fig. 11; results in Fig. 12A–C). *Arrangement.* A context that keeps every component's position and reduces the radial differences between arrangements (components rotated in place about their own ink centroids) is exceeded by the closing context by only +0.01 at an edge length of 5 px, and by +0.24 at 43 px; a fully closed square gives +0.23, on average no more than a nearly closed one (the closing-context CE itself still rises from 43 to 47.5 px in eight of the ten networks, on average from 0.12 to 0.13; it is the closing-minus-in-place contrast that stops growing), so CE grows as the outline approaches completion and the closing contrast gains nothing further, on average, from closing it. The pixel-growth observer shows no such advantage of the closing over the in-place context at 43 px (+0.01; +0.01 at 5 px, and negative at intermediate lengths), so this observer does not account for the large contrast at 43 px, although centroid matching does not equalize the radial histograms exactly. The effect thus tracks how much of the outline is physically present, not gaps to be bridged. Averaged over all conditions, the ten trained networks (nine CNNs and ViT-B/16) score -0.16, mostly a configural inferiority effect, and CE is positive by a one-sample t-test across stimulus sets (nominal $p < 0.05$, uncorrected) in only 22% of network–edge cells with a closing context (0% in place). The pixel-growth observer scores -0.36 on CE itself, so radial ink statistics alone can produce a configural inferiority effect of this size; the arrangement contrast above is what separates the networks from it. *Inducer shape.* Five notched inducers (pac-man, notched square, hexagon, cross and ring) have identical notch edges and hence the same ratio of luminance-specified to total edge length, which predicts the clarity of human illusory boundaries [30], but a network's CE varies across them by a median of 0.16 and changes sign in 5 of 10 networks (ResNet50: +0.03 with pac-men, +0.31 with notched squares). *Inducer size.* Enlarging the inducer disc while keeping the notch edges fixed, which leaves that ratio unchanged, moves CE from -0.18 to -0.11, while the closing context's advantage over the in-place control stays positive (at least +0.02). CE is therefore specific to closing arrangements of nearly complete outlines, but it is not invariant to manipulations that leave that ratio unchanged.

Accuracy under edge removal [38]. This protocol already concludes that CNNs lack closure. Because corners are kept, it can be passed by reading corner angles (see above); because gaps scale with edge length, gap size in patches is confounded with the number of sides; and because, in their released stimulus code, test polygons are smaller than training polygons (radius 60 versus 67 px), the reported drop at 10% removal mixes size change with removal.

4.8 Saliency overlays are often not faithful

Overlays are a common argument that a network "sees" a triangle. We scored each map with a deletion test that removes ink in map order or in random order [25], and call the resulting score faithfulness, in the sense in which Selvaraju et al. [28] use the word (Appendix B). Grad-CAM [28] and class-specific relevance maps [8] have a median faithfulness of +0.020 and are near zero ($|\cdot| < 0.02$) for 7 of 16 networks; occlusion [37] and Shapley values computed over the stimulus's own pieces score +0.051 and +0.043 (Fig. 13). This is separate evidence from the sanity checks of Adebayo et al. [2], which Grad-CAM passed and which test a different property.

5 Discussion

Across 24 networks and four published measures, we find at most a weak illusory-contour signal, consistent across seeds only in ViT-B/16 (Kanizsa accuracy at most 0.39 against 1.00 on complete outlines), and the published illusory-brightness and representational-similarity signatures of closure are reproduced by observers that cannot close contours. What networks do show is integration of nearby, consistently oriented contour pieces: fine-tuned transformers use both the positions and the orientations of fragments, and ImageNet CNNs show a configural effect whose advantage for closing arrangements grows as the outline approaches completion. In CNN and recurrent networks the gaps that halve closure are, with one exception (two counting single-iteration controls), smaller than the measured effective receptive field, so the measured receptive field does not appear to be what limits closure. This places current networks closer to the local association-field account of contour integration [12] than to Gestalt closure: they link pieces that are near and aligned, and do not complete what is absent. A related dissociation has been reported for vision-language models, which give the expected answer to classic illusions but keep it when the inducers are reversed [31]; recognizing an illusory configuration and being susceptible to it are different properties, and our measures target the second. Networks trained to trace contours, the task recurrence is thought to be needed for, share this profile: they link aligned pieces a few pixels beyond their training statistics and label almost none of the missing contour as contour. Recurrence does not change this picture. In some fGRU circuits the gap similarity rises over iterations, but classifiers trained on complete outlines have little reason to use it, and single-iteration controls perform as well in all but one case.

Recommendations. Closure tests should (i) specify gaps in patch or receptive-field units; (ii) remove corners or otherwise prevent corner reading; (iii) include controls that keep the ink, its pixel growth and its orientations while breaking the arrangement; and (iv) report null observers (untrained networks, a filter bank, pixels and a pixel-growth observer) alongside trained ones.

Limitations. ViT-B/16 and DINOv2-S were trained with three seeds and ResNet50 with five, whose fragment and arrangement effects vary by up to 0.03 and 0.05 (s.d.); the others were trained once, so differences of that size between them remain uncertain without repeated training. Published stimuli were recreated from their papers' descriptions rather than original code. We did not test the implicit-prior account of Toosi and Miller [34], or abutting-grating illusions, and we did not collect human data on our stimuli. Pathfinder pretraining [20] of our recurrent circuits, and of a ViT-B/16,

remained at chance after 1–2 h of training per network; networks from a companion project that were trained to trace contours stand in for it (Fig. 9).

Use of AI tools

Code for stimulus generation, training, evaluation and figures, and drafts of this manuscript, were produced with the assistance of a large language model (Claude, Anthropic). All experimental design decisions, results and text were reviewed by the authors.

Code and data

Stimulus generator, training and evaluation code, and per-image results will be released with the final version.

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A Results for every network

Network	complete	frag.	CI frag.	scrambled	conflict: pos.	conflict: ori.	Kanizsa	CI Kan.	Brain-Score*
VGG16 (frozen)	0.99	0.25	+0.14	0.10	0.10	0.00	0.14	+0.02	0.37
ResNet50 (frozen)	0.97	0.41	+0.27	0.12	0.15	0.04	0.20	-0.00	0.38
ViT-B/16 (frozen)	1.00	0.62	+0.43	0.26	0.12	0.16	0.29	+0.10	–
DINOv2-S (frozen)	0.97	0.63	+0.51	0.20	0.02	0.08	0.19	+0.00	0.35
ResNet50	1.00	0.66	+0.53	0.14	0.27	0.15	0.18	+0.04	0.34
ViT-B/16	1.00	0.82	+0.60	0.27	0.05	0.18	0.29	+0.08	–
DINOv2-S	1.00	0.94	+0.79	0.30	0.03	0.33	0.25	+0.10	0.34
Looped ViT-B/16	1.00	0.81	+0.60	0.39	0.04	0.42	0.39	+0.09	–
Looped DINOv2-S	1.00	0.97	+0.80	0.36	0.09	0.30	0.20	+0.03	0.31
fGRU-H	1.00	0.32	+0.13	0.12	0.14	0.12	0.15	-0.00	–
fGRU-H (T=1)	1.00	0.30	+0.18	0.18	0.09	0.11	0.14	+0.00	–
fGRU-H, 1x1 readout	1.00	0.78	+0.63	0.68	0.05	0.70	0.07	-0.00	–
γ-net	1.00	0.58	+0.36	0.41	0.08	0.52	0.04	+0.00	–
γ-net (T=1)	1.00	0.68	+0.50	0.56	0.09	0.53	0.12	+0.04	–
deep fGRU-H D8	1.00	0.54	+0.36	0.11	0.11	0.00	0.16	+0.06	–
deep fGRU-H D8 (T=1)	1.00	0.51	+0.36	0.10	0.10	0.00	0.11	+0.03	–
deep fGRU D4	1.00	0.55	+0.42	0.10	0.16	0.00	0.17	+0.06	–
deep fGRU D4 (T=1)	1.00	0.51	+0.34	0.15	0.10	0.03	0.18	+0.03	–
DN fGRU stack L4	1.00	0.51	+0.29	0.24	0.15	0.12	0.13	-0.01	–
DN fGRU stack L4 (T=1)	1.00	0.49	+0.30	0.24	0.01	0.16	0.09	+0.01	–
DN fGRU stack L8	1.00	0.41	+0.26	0.29	0.12	0.10	0.10	-0.00	–
DN fGRU stack L8 (T=1)	1.00	0.61	+0.40	0.40	0.01	0.21	0.11	-0.00	–
VGG16 (Zhang data)†	0.29	0.19	+0.06	0.12	0.11	0.01	0.17	+0.00	–
ResNet50 (Zhang data)†	0.39	0.28	+0.15	0.13	0.13	0.03	0.10	+0.00	–

Table 1: All networks. complete: accuracy on complete outlines; frag. and CI: accuracy and closure index for corner-removed fragments of the main battery; scrambled and cue conflict: the orientation-control battery (a different set of fragments, so scrambled cannot be subtracted from frag.); Kanizsa: accuracy and closure index; Brain-Score*: mean of public-split V1, V2, V4, IT and behavioural benchmarks run locally [27] (not the leaderboard score; computed for a subset of networks, "–" otherwise). †Trained on the stimuli of Zhang et al. [38] (validation accuracy 0.88 and 0.85 on those stimuli); they transfer poorly to our battery.

Network	family	ERF_{50} or P (px)	g^* (px)	$g^*/(\mathrm{ERF}_{50}$ or $P)$
VGG16 (frozen)	CNN	38	6	0.16
ResNet50	CNN	60	20	0.34
ResNet50 (frozen)	CNN	66	14	0.22
deep fGRU-H D8 (T=1)	T=1 control	16	9	0.58
fGRU-H (T=1)	T=1 control	19	15	0.78
γ-net (T=1)	T=1 control	35	39	1.09
DN fGRU L4 (T=1)	T=1 control	43	11	0.25
DN fGRU L8 (T=1)	T=1 control	60	51	0.84
deep fGRU D4 (T=1)	T=1 control	65	38	0.59
Looped ViT-B/16	ViT (patch)	16	54	3.39
ViT-B/16 (frozen)	ViT (patch)	16	55	3.45
DINOv2-S	ViT (patch)	14	$\geq$56	$\geq$4.00
ViT-B/16	ViT (patch)	16	$\geq$64	$\geq$4.00
DINOv2-S (frozen)	ViT (patch)	14	51	3.64
Looped DINOv2-S	ViT (patch)	14	$\geq$56	$\geq$4.00
deep fGRU-H D8	recurrent	16	10	0.62
fGRU-H, 1×1 readout	recurrent	18	50	2.82
fGRU-H	recurrent	18	15	0.82
γ-net	recurrent	59	29	0.49
deep fGRU D4	recurrent	70	27	0.39
DN fGRU L4	recurrent + DN	58	17	0.30
DN fGRU L8	recurrent + DN	74	44	0.60

Table 2: Critical gap g^* against ERF_{50} (patch size P for transformers) for every network, including the recurrent variants and single-iteration controls not drawn in Fig. 5.

Network	seeds	CI corner	CI mid	arrangement	CI Kanizsa	Kanizsa range
ViT-B/16	3	+0.62 ± 0.03	+0.42 ± 0.04	+0.54 ± 0.05	+0.11 ± 0.03	+0.08 to +0.14
DINOv2-S	3	+0.78 ± 0.01	+0.43 ± 0.01	+0.64 ± 0.02	+0.09 ± 0.07	+0.01 to +0.15
ResNet50	5	+0.52 ± 0.03	+0.23 ± 0.06	+0.51 ± 0.03	+0.02 ± 0.05	-0.05 to +0.07

Table 3: Repeated training seeds: mean ± s.d. of the closure indices and the arrangement effect.

B Implementation details

Training used AdamW with cosine decay and 5% warm-up, label smoothing 0.1 and bf16 autocast (full precision for the DN stacks, whose eight-layer stack falls from 1.00 to 0.44 on complete outlines when evaluated in bf16); backbone learning rates were 10^{-5} (transformers) or 10^{-4} (CNNs and recurrent front ends), and new modules used 10^{-3} (10^{-4} for the looped-transformer injection). fGRU cells used 9×9 horizontal kernels at stride 4 and kernels of 7, 5 and 3 at deeper stages; gate biases followed the chrono initialization [32] and horizontal kernels were orthogonally initialized and then made symmetric between channels; α is kept positive by a softplus, and a top-down fGRU adds its output to the lower layer. Relevance maps followed Chefer et al. [8]: for each block, $\bar{A} = \mathbb{E}_h[(\nabla A \odot A)^+]$ with ∇A the gradient of the class logit, and $R \leftarrow R + \bar{A}R$ from $R = I$, the map being the class-token row of R; unlike attention rollout [1] it is class-specific, and unlike Chefer et al. [9] it uses no layer-wise relevance propagation; for looped transformers, loops are unrolled so each executed block contributes once. Shapley values over pieces were estimated with 16 antithetic permutations over the stimulus's own pieces; the value of a set of pieces is the log probability of the class when only those pieces are drawn, a piece being removed by drawing it in the background

colour.

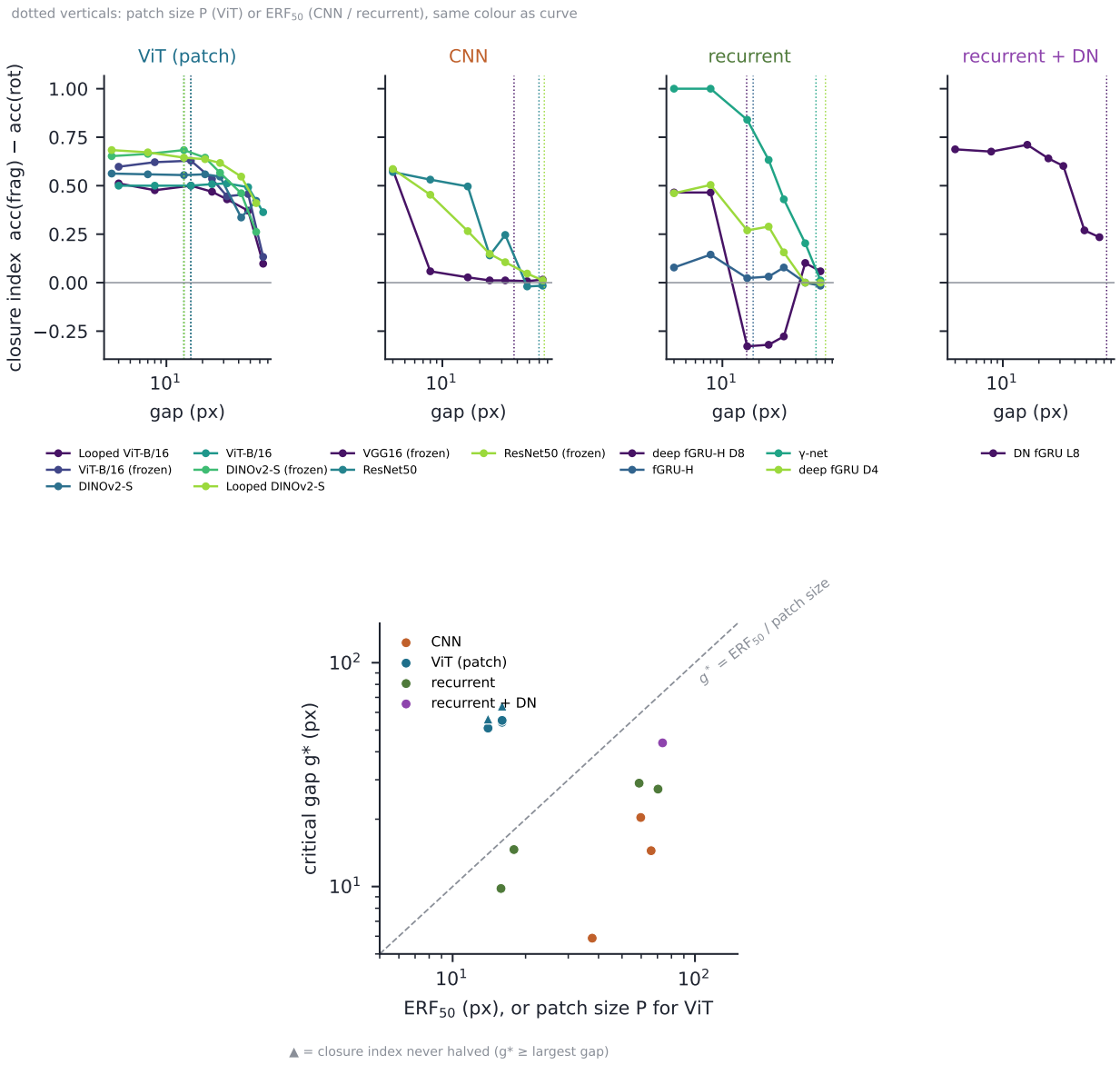


Figure 5: Top: mid-gap closure index against gap size, by family (one network per recurrent family; all networks in Table 2); dotted verticals mark each network's ERF_{50} (patch size for transformers). Bottom: critical gap g^* (closure index halved) against ERF_{50} or patch size; dashed line, equality. Excluding single-iteration controls, all CNN and recurrent networks but one (the fGRU-H with a 1×1 readout) lie below the diagonal; including them, 14 of 16 do.

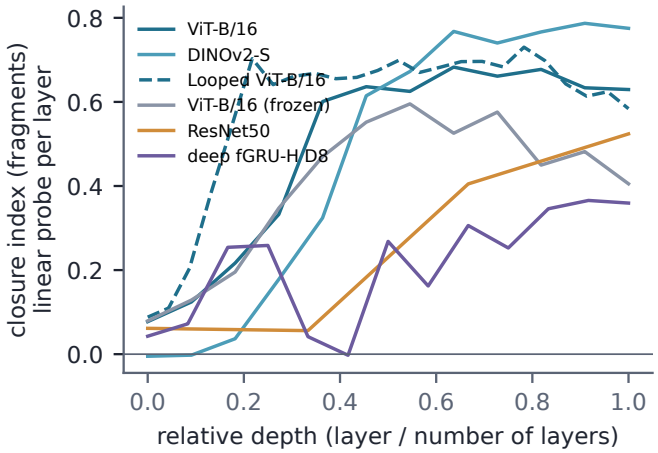


Figure 6: Fragment closure index from a linear probe trained on complete outlines at every layer, against relative depth. For the looped ViT, depth counts executed blocks (loops unrolled).

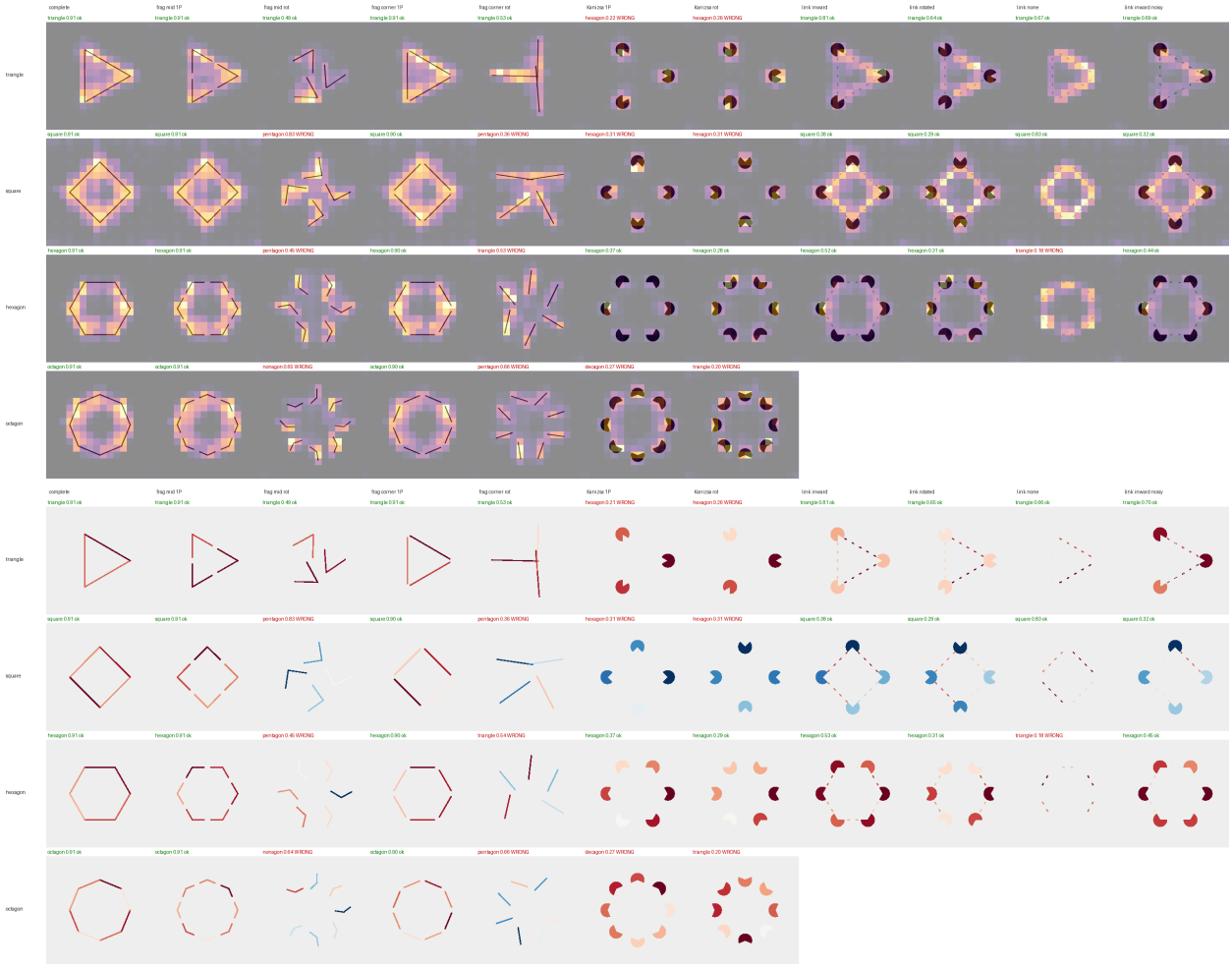


Figure 7: Fine-tuned DINOv2-S. Top: class-specific relevance maps [8] for the top-1 class per condition (columns) and true class (rows); cell titles give the prediction and its probability. Bottom: Shapley values [29] of the stimulus's pieces for the true label (red supports, blue opposes). Illustrative only: these maps are weakly faithful (Section 4.8); the behavioural evidence for integration is in Fig. 2.

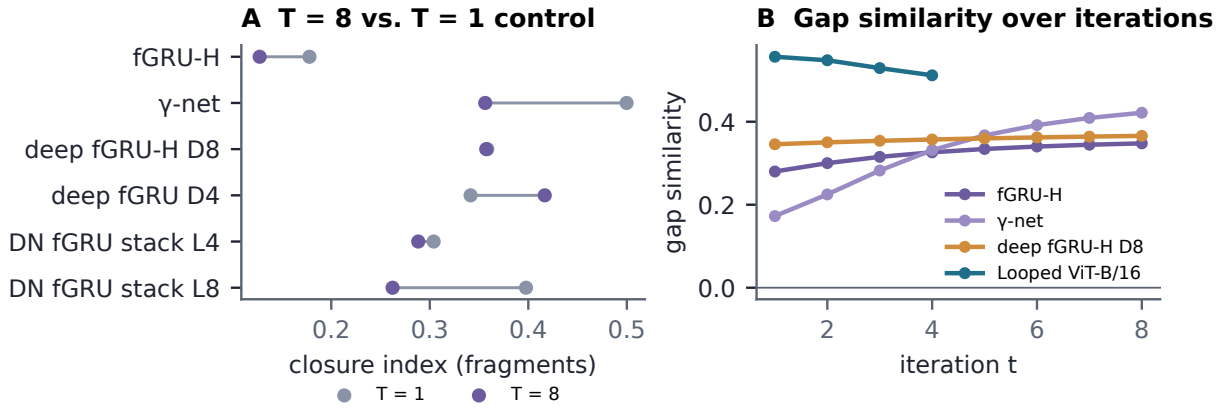


Figure 8: (A) Fragment closure index for the six recurrent circuits with matched controls (T = 8) and their controls trained with a single iteration. (B) Gap similarity in the gap region of the recurrent state (first recurrent layer, one-patch corner-removed fragments) across iterations.

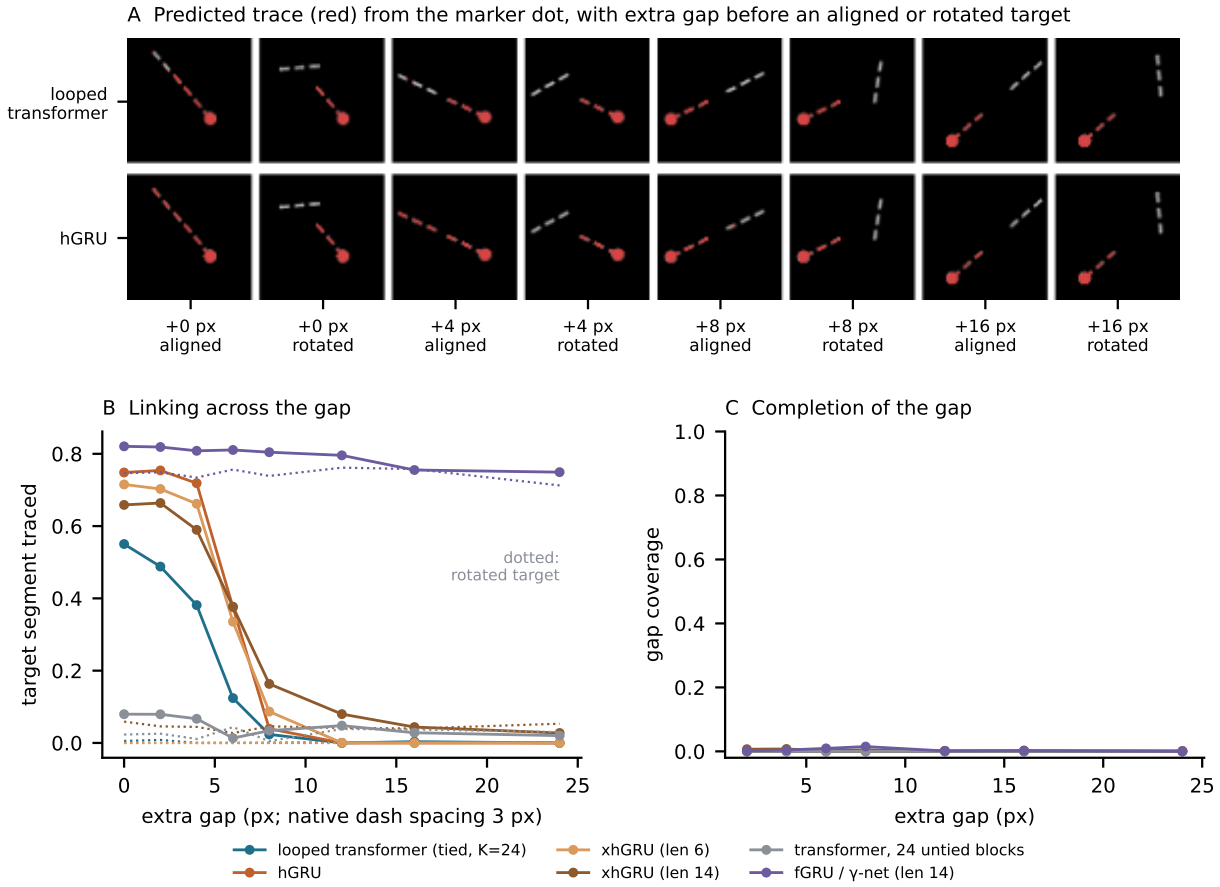


Figure 9: Contour-tracing networks (trained to segment the dashed contour a marker sits on) probed with a marked dashed segment, an extra gap, and a target segment that continues the path or is rotated in place. (A) Predicted traces. (B) Share of the target traced; solid, aligned; dotted, rotated. (C) Share of the missing contour in the gap predicted as contour.

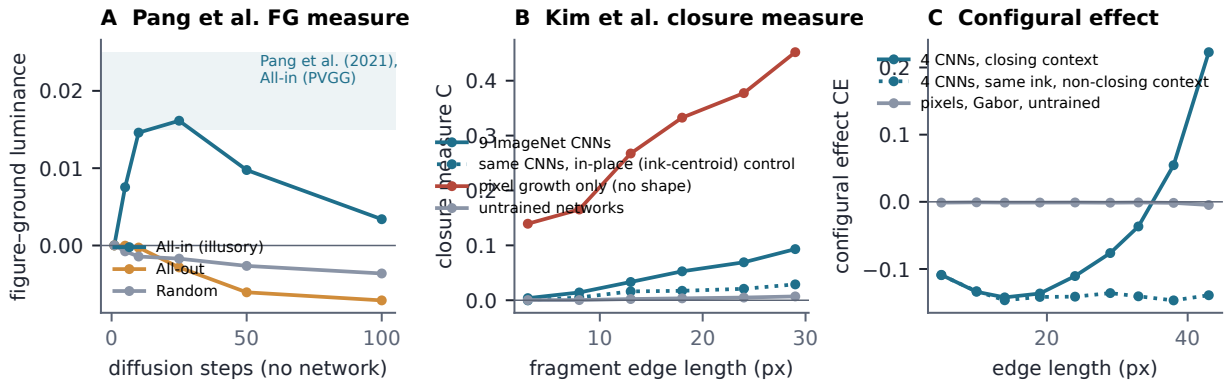


Figure 10: Published measures recomputed on null observers. (A) The figure–ground luminance difference of Pang et al. [24] computed on input images smoothed by diffusion; shaded band, magnitude reported for the All-in condition. (B) The closure measure $\bar{C}$ of Kim et al. [19] and Zhang et al. [39] for nine ImageNet CNNs, for the same CNNs with a disordered control that keeps ink in place, for an observer that only encodes the radial distribution of ink, and for untrained networks. (C) The configural effect (CE) of Zhang et al. [39] with a closing context and with the same components rotated so that no closed shape can form, against pixel, Gabor and untrained observers.

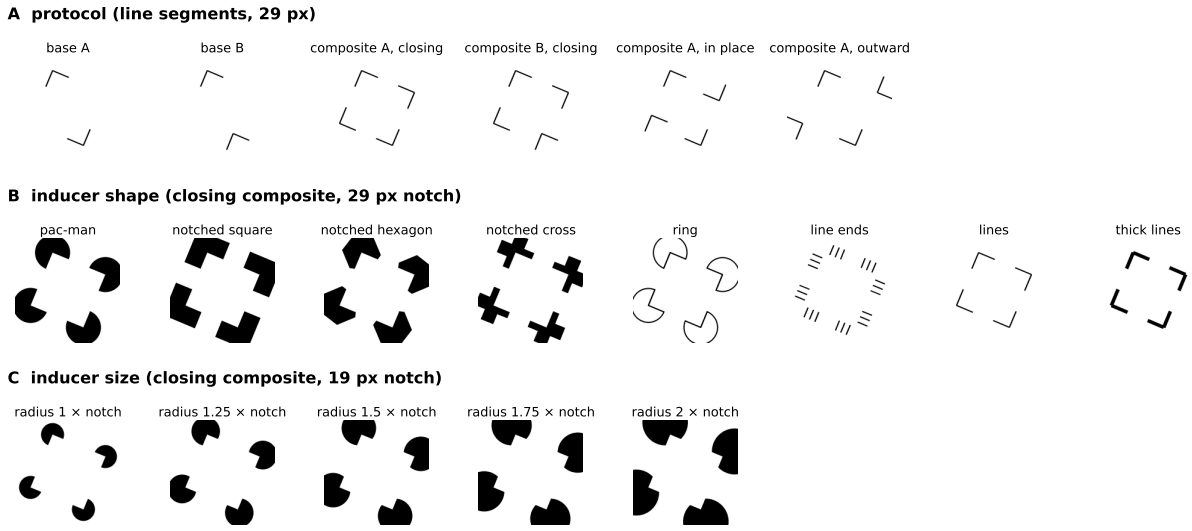


Figure 11: The configural-effect stimuli. (A) The protocol with line segments (edge length 29 px): the base pair differs in the orientation of one corner (flipped about its own ink centroid); the composite pair adds the same two corners, the context, to both. The closing context completes the square; the in-place context rotates each added corner by 90° about its own ink centroid; the outward context rotates it by 180° about its vertex. (B) The eight inducers of the shape audit, each specifying the same corner of the same square with the closing context (29 px notch); the first five have identical notch edges. (C) The size audit: pac-man discs enlarged at a fixed 19 px notch, so the notch edges and the implied square are unchanged.

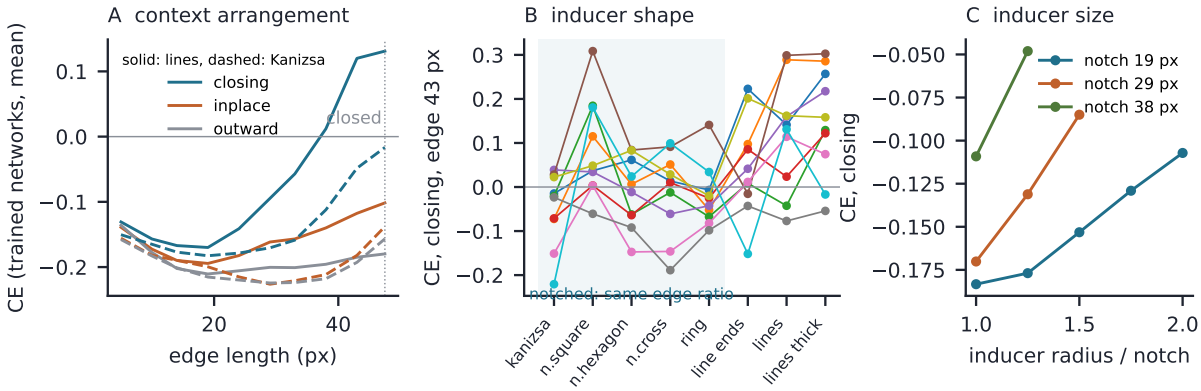


Figure 12: (A) Configural effect for the ten trained networks (nine CNNs and ViT-B/16) with closing, in-place (component-centroid-matched, non-closing) and outward contexts against edge length; 47.5 px is a closed square. (B) CE per network across inducer shapes; shaded, notched inducers with identical notch edges (same specified-to-total edge ratio). (C) CE against inducer radius at a fixed notch.

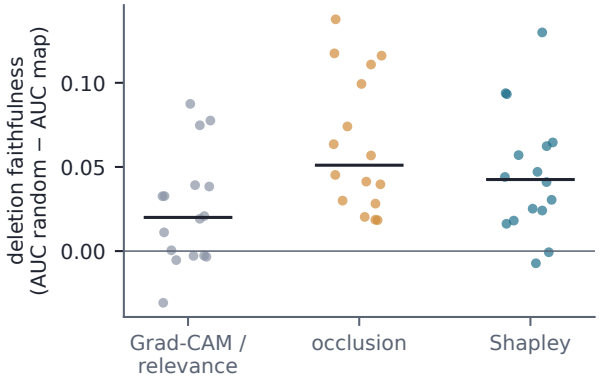


Figure 13: Deletion faithfulness of three attribution methods across networks (one point per network; bar, median).